AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MST224
Mathematical Methods

Midterm Examination (MTA) - Makeup
Spring Semester 2024/2025

Date: 22 /April/2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2H

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

[6×2 marks] Choose the correct answer:

1. The value of m so that $\mathbf{a} = 2\mathbf{i} + m\mathbf{j} + \mathbf{k}$ and $\mathbf{b} = 4\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ are perpendicular is:
   a) $m = 1$.
   b) $m = 3$.
   c) $m = \frac{1}{2}$.
   d) None of the above.

2. The general solution of the differential equation $y''' - 6y'' + 11y' - 6y = 0$, is:
   a) $C_1e^x(C_2\cos 2x + C_3\sin 3x)$.
   b) $e^x(C_1x^2 + C_2x + C_3)$.
   c) $C_1e^x + C_2e^{2x} + C_3e^{3x}$.
   d) None of the above

3. Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then $\mathbf{A}^{-1} =$
   a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
   b) $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
   c) $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$
   d) None of the above.

4. The magnitude of the vector $\mathbf{a} = \mathbf{b} - 3\mathbf{c}$, where $\mathbf{b} = 3\mathbf{i} + 4\mathbf{j} - \mathbf{k}$, and $\mathbf{c} = -\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$, is:
   a) $|\mathbf{a}| = \sqrt{53}$
   b) $|\mathbf{a}| = 3\sqrt{21}$
   c) $|\mathbf{a}| = 13$
   d) None of the above.

5. Which of the following vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel?
   a) $\mathbf{a} = (2,3,4)$ and $\mathbf{b} = (1,-1,-3)$
   b) $\mathbf{a} = (3,5,-2)$ and $\mathbf{b} = (3,-1,-2)$.
   c) $\mathbf{a} = (2,3,1)$ and $\mathbf{b} = (2,-1,-1)$
   d) None of the above.

6. If the position of a particle at time t is given by $\mathbf{r}(t) = 3(1 - \cos 2t)\,\mathbf{i} + (2\sin 2t - 4t)\,\mathbf{j} + \frac{8}{3}t^3\,\mathbf{k}$, then its acceleration is:
   a) $\mathbf{a}(t) = 12\cos 2t\,\mathbf{i} - 8\sin 2t\,\mathbf{j} + 16t\,\mathbf{k}$
   b) $\mathbf{a}(t) = 6\sin 2t\,\mathbf{i} + (4\cos 2t - 4)\,\mathbf{j} + 8t^2\,\mathbf{k}$
   c) $\mathbf{a}(t) = \left(3t - \frac{3}{2}\sin 2t\right)\mathbf{i} + (\cos 2t - 2t^2)\mathbf{j} + \frac{2}{3}t^4\,\mathbf{k}$
   d) None of the above.

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PART 2

1. **Find the general solution of the following differential equations:**
   i. **[4 marks]** $t^2y' + ty = 2.$
   ii. **[4 marks]** $xe^{-y}\sin x - yy' = 0$
   iii. **[4 marks]** $\frac{1}{4}y'' + y' + y = x^2 - 2x.$

2. Consider the following system of differential equations:

$$\begin{cases} \dot{x} = y, \\ \dot{y} = -x \end{cases}$$

   i. **[4 marks]** Find the general solution of this system.
   ii. **[2 marks]** Obtain the real-valued solution.

End of questions

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